Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
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CH615S – INORGANIC CHEMISTRY - IV

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Which of the following metals in enzymes is not involved in the nitrogen cycle?
   a) Fe b) Ni c) Cu d) Mo

2. Complete the reaction $_{7}N^{14} + _{2}He^{4} \longrightarrow ____ + _{1}H^{1}$
   a) $_{8}O^{17}$ b) $_{10}Ne^{18}$ c) $_{9}F^{17}$ d) None of these

3. The common oxidation state of lanthanides is _____.
   a) +2 b) +3 c) +4 d) +5

4. The oxidation state of zinc in carbonic anhydrase is:
   a) 0 b) +1 c) +2 d) +3

5. The elements with atomic number greater than 103 are called _____
   a) Noble metals b) Rare metals
   c) Super-heavy elements d) All the above

6. The hapticity of cyclopentadienyl ligands in a stable ferrocene is:
   a) 2 b) 4 c) 5 d) 6

7. In oxo process_____ catalyst is used.
   a) Platinum b) Nickel c) Palladium d) Cobalt

8. The dioxygen in oxy-hemoglobin is in_____ oxidation state
   a) 0 b) + 2 c) -1 d) +3

9. The shift reagent used in NMR technique is_____
   a) $Eu(fod)_3$ b) $AlCl_3$ c) $TiCl_4$ d) $Al(Et)_3$

10. The _____ of a nucleus represents the mass of the energy binding the nucleus
    a) Mass defect b) Nucleons
    c) Nuclear force d) None of these

II. Fill up the blanks

11. _____ strong inhibitor of O_2 binding by myoglobin and haemoglobin.

12. The neptunium series ends with_____.

13. The basic principle used in the hydrogen bomb is_____.

14. The metal ion present in vitamin B_{12} is _____.

15. Repee's catalyst is _____

III. Match the following

16. Atom bomb - ^{40}Ca&^{40}Ar

17. Probability of a nuclear reaction - $PtCl_4$

18. Isotones - Nuclear fission

19. Wacker's process - Mo

20. Nitrogen fixation - Q-value

- ^{36}S&^{37}Cl

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Define packing fraction with example.

22. What are isobars?

23. Where the sun and stars get their energy from?

24. What do you meant by Q-value?

25. What is the role of Mg in Na-K pump?

26. How hemoglobin differs from myoglobin?

27. What is hapicity?

28. List out the reasons of lanthanide contraction?

29. Write about the oxidation states of Cerium, Germanium.

30. What is Wilkinson's catalyst?

31. Write the molecular formula of the vaska complex.

32. Give detailsof the term polymer bound catalyst.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. i) How will you detect and measure the radioactivity of an element? **(5+2)**
    ii) Define average-life period

34. i) Describe the importance of N/P ratio **(5+2)**
    ii) Define group displacement law.

35. i) Classify the types of nuclear reactions.
    ii) Explain the different types of nuclear reactions
    iii) What is fast breed reactor? **(2+3+2)**

36. i) What is cyclotron?
    ii) How cyclotron works explain with a diagram?
    iii) What are the uses of accelerator? **(3+3+1)**

37. i) Draw the structure of chlorophyll
    ii) Mg^{2+} is the metal ion present in chlorophyll, list out its role in photosynthesis. **(3+4)**

38. Draw the structure of
    i) Ti $(C_5H_5)_4$
    ii) $Fe_2(CO)_9$
    iii) Write a note on oxidation state of lanthanides. **(2+2+3)**

39. Explain the oxidation of olefins to aldehydes and ketones by wacker's process. **(7)**

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